TASK 1 — SEQUENTIAL HANDWRITTEN KANNADA DIGIT CLASSIFICATION

Report based on the executed Colab notebook and its recorded outputs

Deep Learning — Assignment 4

1. Introduction

Task 1 formulates handwritten Kannada numeral recognition as a sequence-classification problem. Each 28 × 28 grayscale image is presented to a recurrent network one row at a time, producing a sequence of 28 feature vectors with 28 values per timestep. The notebook compares Vanilla RNN, LSTM, and GRU architectures, then explores bidirectional recurrent processing and temporal attention.

2. Dataset and preprocessing

The executed notebook loaded the labelled Kannada-MNIST training file, containing 60,000 samples and 785 columns: one label column and 784 pixel columns. All ten classes (0–9) were present. Pixel values were converted to floating point and normalized by dividing by 255, placing them in [0, 1]. The 784 pixel values were reshaped into (28 timesteps, 28 features) per image.

Split	Samples	Percentage	Samples per class
Training	42,000	70%	4,200
Validation	9,000	15%	900
Held-out test	9,000	15%	900
Total	60,000	100%	6,000

The split was stratified and used a fixed random seed, so each class was represented equally in the three subsets. Model selection was based on validation accuracy. Because test scores were also recorded for multiple configurations, they are reported for comparison and should be interpreted as held-out estimates rather than as a completely untouched final evaluation.

3. Sequential input representation

For an image $X \in R^{28 \times 28}$, timestep t receives the t-th image row as $x_t \in R^{28}$. The input batch shape reported by the notebook was (128, 28, 28), corresponding to batch size, sequence length, and feature dimension. This matches the required row-by-row formulation.

4. Model architectures and training setup

The baseline search tested six hidden-size layouts for each recurrent cell type: [32], [64], [128], [32, 32], [64, 64], and [32, 64]. This produced 18 baseline experiments. Six additional experiments tested bidirectional networks, temporal attention, and their combination. The output layer predicts one of ten digit classes.

Setting	Value
Recurrent cells	Vanilla RNN, LSTM, GRU
Input dimensions	28 timesteps × 28 features
Output classes	10
Optimizer	Adam
Learning rate	0.001
Batch size	128
Convergence threshold	$\lvert\bar{L}(k) - \bar{L}(k-1)\rvert < 10^{-4}$
Maximum epochs	100 (safety cap; convergence checked each epoch)
Model selection	Highest validation accuracy

The stopping rule compares average training loss between consecutive epochs. The 100-epoch limit is a safety cap, not the convergence criterion. The saved results show 22 of 24 runs met the threshold; RNN_64-64 and RNN_64_Bi reached the cap without convergence.

5. Digit visualizations

The notebook displays five training examples from each of the ten classes, giving 50 sample images. These visualizations provide a qualitative check that all classes are represented and that the raw images are interpreted as 28 × 28 grayscale numerals.

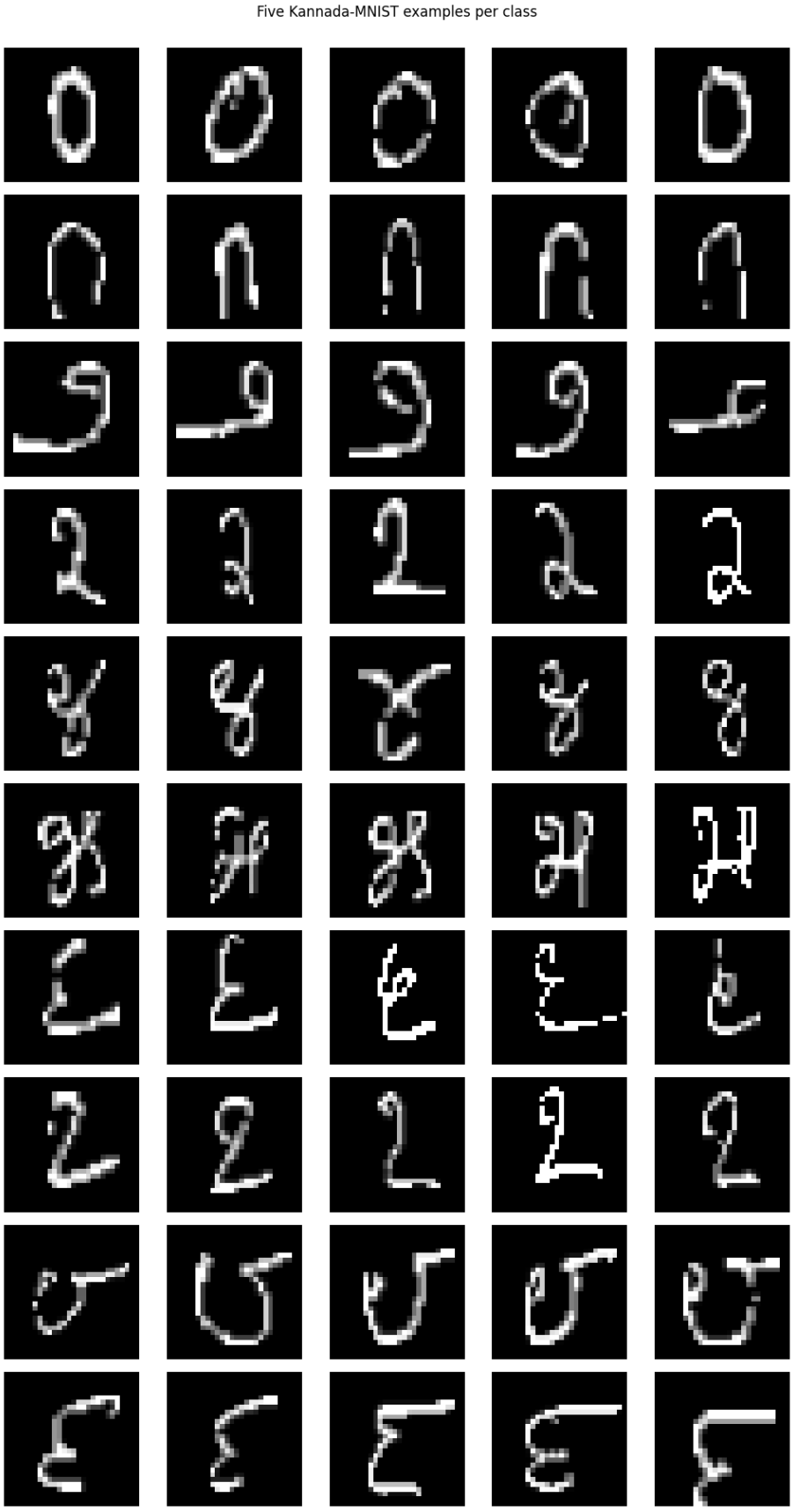


Figure 1. Five sample images from each Kannada digit class (0–9).

6. Training-loss curves

Training-loss curves were generated for all 24 explored configurations. The plots allow comparison of convergence speed and loss behaviour across recurrent cell types, hidden widths, depth, and bonus architectures. They should be interpreted together with the final accuracy and convergence flags.

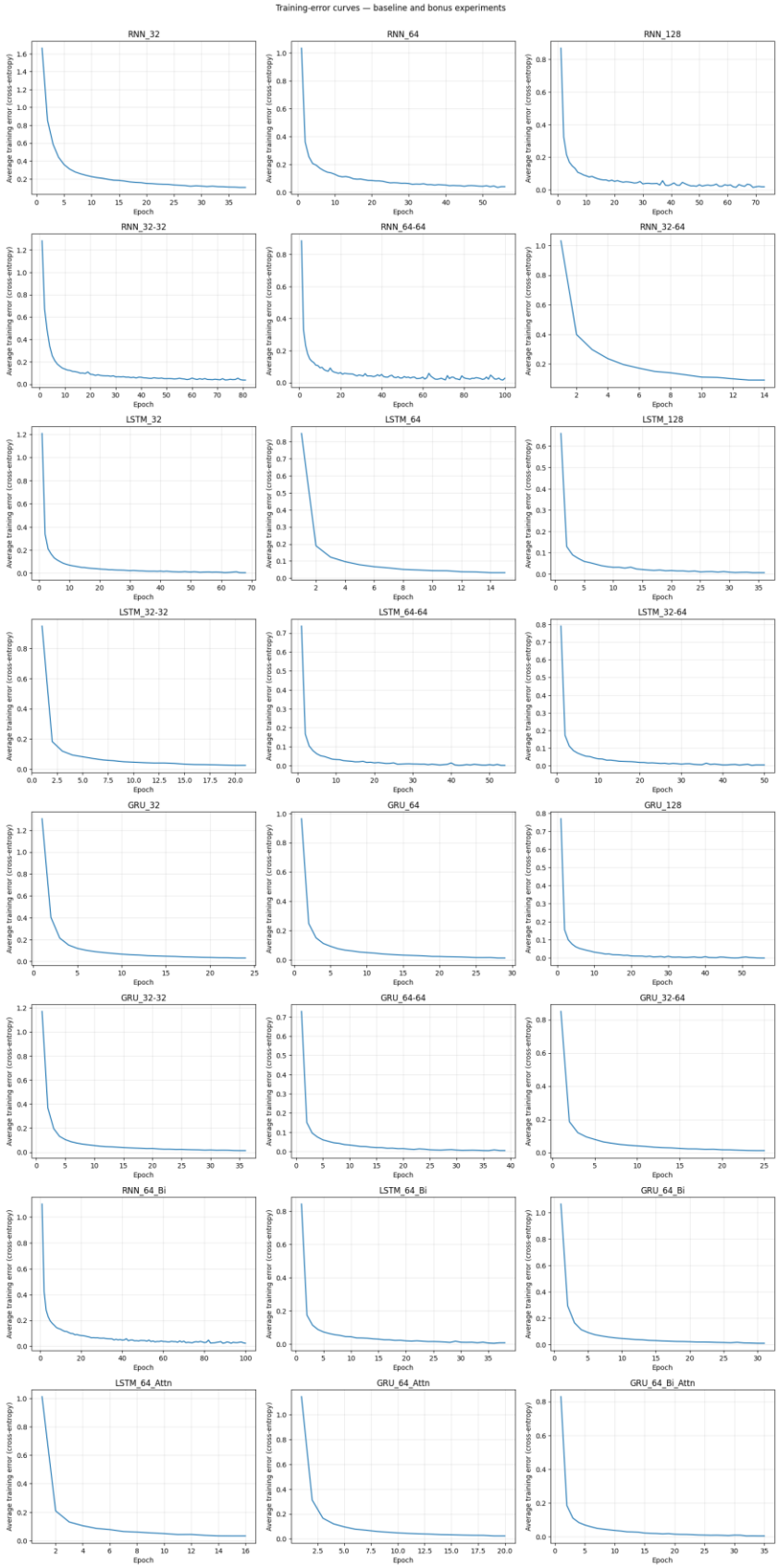


Figure 2. Average training loss versus epoch for all 24 experiments.

7. Results across all configurations

Table 3 reports the recorded training, validation, and held-out test accuracies. Configurations are sorted by validation accuracy, which is the selection criterion used in the notebook. Percentages are rounded to two decimal places.

Run	Group	Hidden sizes	Epochs	Converged	Train	Validation	Test
LSTM_64-64	Baseline	[64, 64]	54	Yes	99.99%	99.33%	99.41%
GRU_128	Baseline	[128]	56	Yes	100.00%	99.30%	99.47%
LSTM_128	Baseline	[128]	36	Yes	99.88%	99.23%	99.40%
LSTM_32-64	Baseline	[32, 64]	50	Yes	99.96%	99.22%	99.41%
GRU_32-64	Baseline	[32, 64]	25	Yes	99.66%	99.10%	99.10%
GRU_64_Bi_Attn	Bidirectional + attention	[64]	35	Yes	99.91%	99.07%	99.17%
GRU_64-64	Baseline	[64, 64]	39	Yes	99.92%	98.99%	99.30%
LSTM_64	Baseline	[64]	15	Yes	99.30%	98.92%	99.01%
GRU_64_Bi	Bidirectional	[64]	31	Yes	99.65%	98.90%	99.06%
LSTM_64_Bi	Bidirectional	[64]	38	Yes	99.80%	98.90%	99.07%
GRU_32-32	Baseline	[32, 32]	36	Yes	99.64%	98.90%	98.98%
LSTM_32	Baseline	[32]	68	Yes	99.91%	98.87%	99.01%
GRU_64	Baseline	[64]	29	Yes	99.59%	98.84%	98.99%
LSTM_32-32	Baseline	[32, 32]	21	Yes	99.30%	98.74%	98.91%
LSTM_64_Attn	Temporal attention	[64]	16	Yes	99.30%	98.74%	98.77%
GRU_64_Attn	Temporal attention	[64]	20	Yes	99.34%	98.64%	98.79%
RNN_128	Baseline	[128]	73	Yes	99.49%	98.52%	98.47%
GRU_32	Baseline	[32]	24	Yes	99.08%	98.39%	98.62%
RNN_64-64	Baseline	[64, 64]	100	No	99.13%	98.36%	98.56%
RNN_32-32	Baseline	[32, 32]	81	Yes	99.11%	98.29%	98.38%
RNN_64	Baseline	[64]	56	Yes	99.07%	98.21%	98.34%
RNN_64_Bi	Bidirectional	[64]	100	No	99.17%	98.13%	98.31%
RNN_32-64	Baseline	[32, 64]	14	Yes	97.53%	97.10%	97.29%
RNN_32	Baseline	[32]	38	Yes	96.76%	96.40%	96.41%

8. Baseline family comparison

For each baseline cell type, the configuration with the highest validation accuracy was selected for the family-level comparison. GRU [128] had the highest test accuracy among these family winners, while LSTM [64, 64] had the highest validation accuracy and was selected as the overall best model.

Cell type	Best baseline run	Train accuracy	Validation accuracy	Test accuracy	Epochs	Converged
RNN	RNN_128	99.49%	98.52%	98.47%	73	Yes
LSTM	LSTM_64-64	99.99%	99.33%	99.41%	54	Yes
GRU	GRU_128	100.00%	99.30%	99.47%	56	Yes

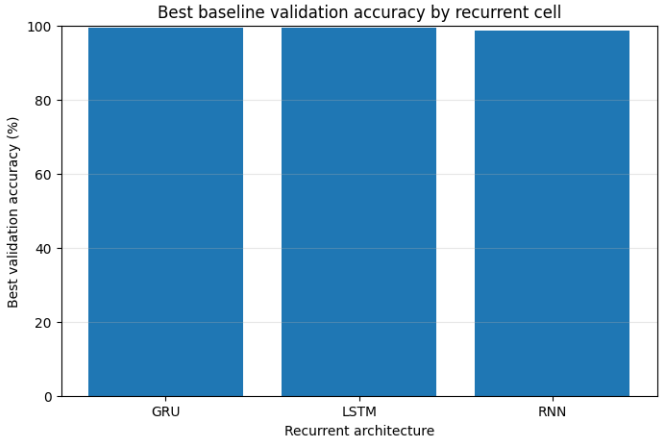


Figure 3. Best validation accuracy for the best baseline configuration of each recurrent cell type.

9. Best configuration and held-out evaluation

The notebook selected LSTM_64-64 by validation accuracy. It is a two-layer unidirectional LSTM with hidden sizes [64, 64]. The saved run converged after 54 epochs.

Metric	Recorded value
Training accuracy	99.9905%
Validation accuracy	99.3333%
Held-out test accuracy	99.4111%
Final training loss	0.001807
Epochs	54
Convergence	Met threshold

The test accuracy corresponds to 8,947 correct predictions out of 9,000 held-out examples. The classification report in the notebook gives a macro-average precision, recall, and F1-score of approximately 99.41%.

10. Complete confusion matrix

Rows correspond to true labels and columns to predicted labels. Each row contains 900 test samples. The diagonal contains correct predictions; off-diagonal entries show the misclassifications.

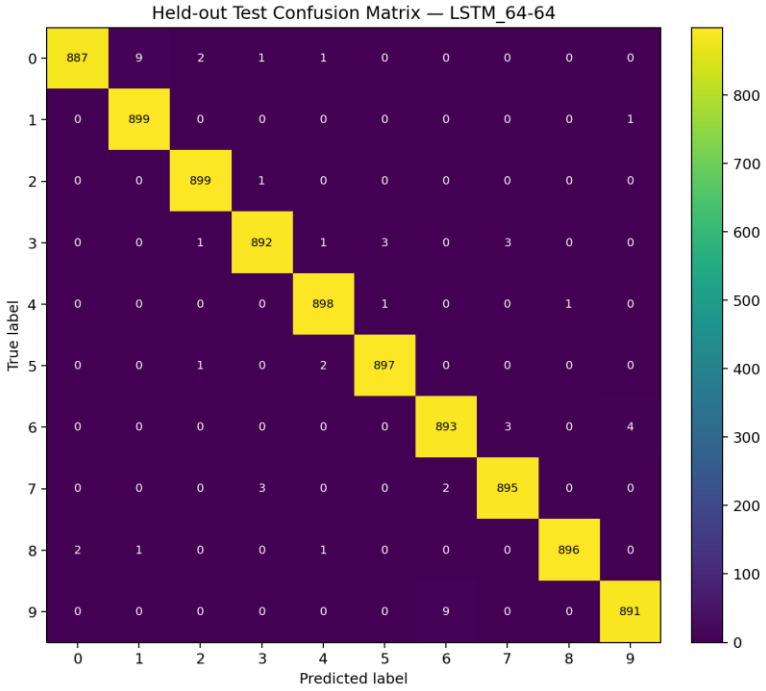


Figure 4. Numeric confusion matrix for the selected LSTM_64-64 model.

True\Pred	0	1	2	3	4	5	6	7	8	9
0	887	9	2	1	1	0	0	0	0	0
1	0	899	0	0	0	0	0	0	0	1
2	0	0	899	1	0	0	0	0	0	0
3	0	0	1	892	1	3	0	3	0	0
4	0	0	0	0	898	1	0	0	1	0
5	0	0	1	0	2	897	0	0	0	0
6	0	0	0	0	0	0	893	3	0	4
7	0	0	0	3	0	0	2	895	0	0
8	2	1	0	0	1	0	0	0	896	0
9	0	0	0	0	0	0	9	0	0	891

The most noticeable off-diagonal entries are true class 0 predicted as class 1 (9 cases), and true class 9 predicted as class 6 (9 cases). Most other class-wise errors are small, consistent with the high diagonal counts.

11. Bonus experiments

Run	Bonus type	Validation	Test	Epochs	Converged
GRU_64_Bi_Attn	Bidirectional + attention	99.07%	99.17%	35	Yes
GRU_64_Bi	Bidirectional	98.90%	99.06%	31	Yes
LSTM_64_Bi	Bidirectional	98.90%	99.07%	38	Yes
LSTM_64_Attn	Temporal attention	98.74%	98.77%	16	Yes
GRU_64_Attn	Temporal attention	98.64%	98.79%	20	Yes

Run	Bonus type	Validation	Test	Epochs	Converged
RNN_64_Bi	Bidirectional	98.13%	98.31%	100	No

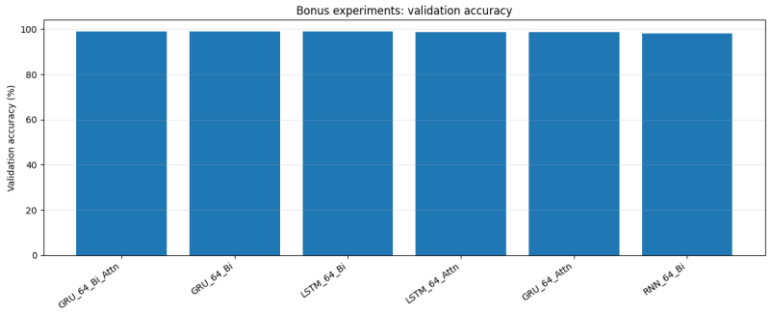


Figure 5. Validation accuracy for the bonus architectures.

Among the saved bonus runs, GRU_64_Bi_Attn achieved 99.07% validation and 99.17% test accuracy. These results are exploratory and did not surpass the selected baseline LSTM on validation accuracy. Implementation limitation: the non-attention bidirectional models use the output at the final timestep as the classifier representation, rather than explicitly concatenating the final forward and backward states. Consequently, the non-attention bidirectional results should be interpreted cautiously.

12. Gradient-flow observations

The notebook records recurrent gradient norms during training and plots them for all experiments. The recorded final gradient norm for the selected LSTM_64-64 run was approximately 0.0815. The baseline family comparison also shows that the RNN family generally required more epochs for its strongest validation configuration than the selected LSTM and GRU configurations. Gradient norms are diagnostics, not proof by themselves of vanishing or exploding gradients; conclusions should be limited to the observed runs.

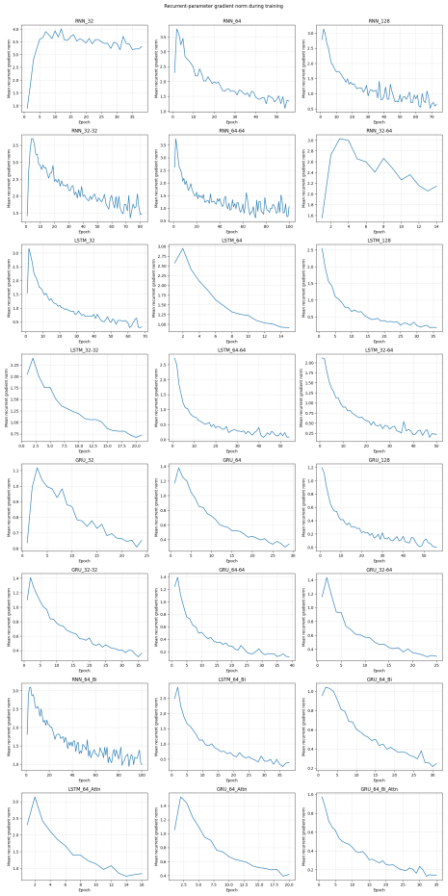


Figure 6. Recurrent gradient-norm histories for the explored models.

13. Temporal attention visualization

The notebook includes a visualization of the temporal attention weights for a sample input. Attention assigns different weights to different image rows, offering a view of which timesteps contribute more strongly to the final pooled representation.

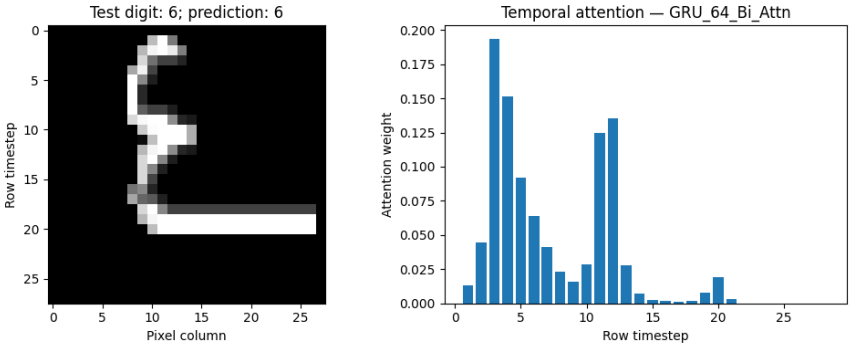


Figure 7. Example temporal attention weights and input visualization.

14. Discussion

All three recurrent cell types performed strongly on the row-wise Kannada-MNIST representation. The selected LSTM achieved 99.41% held-out test accuracy, while the best-validation GRU baseline achieved 99.47% test accuracy. The latter has a slightly higher test score, but model selection was based on validation accuracy as intended; the test set was not used to choose the overall model.

The Vanilla RNN configurations generally showed lower accuracy than the strongest LSTM and GRU configurations in these runs, particularly for smaller or uneven two-layer hidden-size layouts. Increasing hidden size helped some models, but depth or width alone did not guarantee better results. The experiments also show that convergence speed differs substantially by architecture: several LSTM and GRU runs stopped in fewer epochs, whereas two RNN runs reached the safety cap.

The bonus bidirectional and attention experiments were not uniformly better than the baseline models. Their results should be treated as exploratory, especially for non-attention bidirectional models due to the representation limitation noted above. The gradient plots complement the accuracy and loss curves but do not establish causal explanations for performance differences.

15. Conclusion

Task 1 implemented a complete row-by-row recurrent classification pipeline for ten Kannada digit classes. The notebook normalized the pixels, created stratified train/validation/test splits, trained 24 RNN/LSTM/GRU configurations, recorded convergence behaviour, and produced training curves, comparison tables, digit examples, gradient diagnostics, a confusion matrix, and attention visualization. The selected validation-based model was LSTM_64-64, which achieved 99.41% test accuracy. The results support recurrent networks as effective models for this sequential image representation, while also showing that architectural changes and bonus mechanisms should be evaluated empirically.